

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is  $-0.5$ . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

### 1. Perceptron Model

**Given:**

- Weight for Study Hours,  $w_1 = 0.6$
- Weight for Attendance,  $w_2 = 0.4$
- Bias,  $b = -0.5$
- Study Hours,  $x_1 = 2$
- Attendance,  $x_2 = 1$

#### Step 1: Calculate Net Input

**Net Input = 1.1**

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#### Step 2: Apply Step Activation Function

Step Activation Function:

- If **Net Input**  $\geq 0$ , Output = 1
- If **Net Input**  $< 0$ , Output = 0

Since,

[  
1.1  
]

**Output = 1**

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### Step 3: Conclusion

- **Output = 1**
- **Interpretation:** The student **Passes**.

### Final Answer:

- **Net Input = 1.1**
- **Perceptron Output = 1**
- **Result: The student passes.**

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

## 2. Backpropagation Learning Rule

### Given:

- Initial Weight,  $w = 0.5$
- Learning Rate,  $\eta = 0.1$
- Input,  $x = 2$
- Actual Output,  $y = 0.6$
- Target Output,  $t = 1$

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### Step 1: Calculate the Error

**Error = Target Output – Actual Output**

$$\begin{aligned} \text{Error} &= 1 - 0.6 \\ &= 0.4 \end{aligned}$$

**Error = 0.4**

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### Step 2: Calculate the Weight Update (Delta Rule)

### Delta Rule Formula:

$$\Delta w = \eta \times \text{Error} \times x$$

Substitute the values:

$$\Delta w = 0.1 \times 0.4 \times 2$$

$$\Delta w = 0.08$$

**Weight Update = 0.08**

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### Step 3: Calculate the New Weight

$$\text{New Weight} = \text{Old Weight} + \Delta w$$

$$= 0.5 + 0.08$$

$$= 0.58$$

**New Updated Weight = 0.58**

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### Final Answer

- **Error = 0.4**
- **Weight Update ( $\Delta w$ ) = 0.08**
- **New Updated Weight = 0.58**

**Conclusion:** After one iteration of backpropagation using the delta rule, the neuron's weight is updated from **0.5** to **0.58**.